

# Titanic Dataset EDA Insights Summary

## 1. Gender & Survival

Females had a much higher survival rate compared to males. Gender is a strong predictor of survival.

## 2. Class & Survival

Passengers in 1st class had higher chances of survival. Lower classes (especially 3rd) had lower survival rates.

## 3. Age Distribution

Most passengers were between 20-40 years old. Children had a higher survival rate compared to adults.

## 4. Fare Distribution

Fares were right-skewed. Passengers who paid higher fares tended to survive more.

## 5. Missing Values

Age and Cabin had missing values. These need to be filled or handled before modeling.

## 6. Correlation

Pclass and Fare showed notable correlation with survival. Age and Fare had weak correlation with each other.

## 7. Outliers

Some extreme Fare values were found-may need transformation or capping.

## 8. Multicollinearity

No serious multicollinearity detected among major numeric features.